

Question	Marking guidance	Mark	Comments
04.1	$I_1 = 0.80 \text{ A}$	1	Accept 1 sig. fig. in this case
04.2	$R = 1.9 \Omega$	1	
04.3	effect : Reading of A_1 decreases	1	Max 3
	reason: because the total circuit resistance has increased	1	(reason marks are dependent on effect marks)
	effect: Reading of A_2 remains constant	1	
	reason: The p.d. and resistance for this branch are constant.	1	Accept lower part of circuit is unaffected by the switch
	Alternative:		
	$A_1 = A_2$	1	Max 3
	Total circuit resistance = $2R$	1	
	Current = 0.40 A	1	
	So A_1 decreases (A_2 stays the same)	1	

Question	Marking guidance	Mark	Comments
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Question	Marking guidance	Mark	Comments
05.1	$E = 13.60 - 1.51 = 12.09 \text{ eV}$	1	
	Multiplies by 1.6×10^{-19} OR divides by 6.63×10^{-34}	1	
	$f = 2.9 \times 10^{15} \text{ Hz}$	1	
05.2	13.6 eV	1	Do not accept -13.6 eV
05.3	(All of the) photon's energy is absorbed or photon disappears	1	Max 3
	Atom is ionised	1	
	4.8 eV surplus energy	1	
	Plausible destination of surplus energy (eg K.E. of emitted e^- , recoil of nucleus).	1	

Question	Marking guidance	Mark	Comments
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11	B
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15	C
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	$2.3 \times 10^8 \text{ (m s}^{-1}\text{)} \checkmark$	v	
03	any four: V_1 decreases $\checkmark$ because voltage is shared in the ratio of the resistances $\checkmark$ and the variable resistor receives a larger (share of the) voltage $\checkmark$ V_2 stays the same $\checkmark$ because the parallel branches are independent or V_2 is still emf $\checkmark$	For V_1: Same current through R and var R, $\checkmark$ $V = IR$ therefore larger R gets more V $\checkmark$ OR: the current (in that branch) decreases $\checkmark$ Since R is fixed, and $V=IR$, V decreases $\checkmark$ Incorrect conclusion negates explanation marks	4

04.1	Any two.	For MP1 allow a description of diffraction	
	Diffraction at (each) slit ✓ Interference (between light from different slits) ✓ Idea that different wavelengths have maxima at different angles ✓	For MP2 accept discussion of phase difference at maxima / minima	2

Question	Answers	Additional comments/Guidelines	Mark	ID details
04.1	Use of $I = \frac{V}{R}$ or total resistance is $0.015 + 0.0076 (\Omega)$ ✓ 540 (A) ✓	Alternative for MP1: determination of V across the external resistance (8.1 V) Allow 530 A	2	E

Question	Answers	Additional comments/Guidelines	Mark	ID details
04.2	Use of $P = I^2 R$ ✓ 4370 or 4400 (W) ecf ✓	Allow ecf for candidate's $0.041^2 \times 0.015$ Credit alternative routes Calculation of heat in external resistor (2220 W) gets 1 MAX	2	E
Total			4	

Question	Answers	Additional comments/Guidelines	Mark	ID details
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Question	Marking guidance	Mark	Comments
10.1	Uses $R = \frac{\rho l}{A}$ ✓ Uses $A = wt$ leading to 4.45×10^{-5} (m) to at least 2 sf ✓	2	By substitution or manipulation to make A subject. No POT penalty. Expect to see $A = 3.34 \times 10^{-7} \text{ m}^2$

Question	Marking guidance	Mark	Comments
10.2	Voltmeter reading decreases (as the level in the fuel tank falls)✓ as resistance in parallel with voltmeter decreases OR resistance not in parallel increases✓	2	Condone 'resistance voltmeter measures' for resistance in parallel

Question	Marking guidance	Mark	Comments
10.3	Use of $P = \frac{V^2}{R}$ ✓ 0.25(3) (W) ✓	2	Allow MP1 for use of $V = IR$, and using $P = IV$ Calculator value is 0.252631579

Question	Marking guidance	Mark	Comments
10.4	No change in operation (as the pd is divided in the same way) ✓	1	Allow idea that current is less OR power/energy consumption is less OR resistor does not get (so) hot.
Total		7	

Question	Key	Answer
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Question	Answers	Additional comments/Guidelines	Mark	AO
08.1	Correctly determine resistance of parallel combination or adds their resistance of parallel combination to the $2\ \Omega$ ✓ Answer $R = 3.2\ \Omega$ ✓		2	2 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
08.2	Start with a version of $\varepsilon = I(R + r)$ At least one intermediate step to give $\frac{1}{I} = \frac{R}{\varepsilon} + \frac{r}{\varepsilon}$ ✓		1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
08.3	Uses data from graph from at least half the range ✓ Use of valid method to evaluate $\varepsilon (= \frac{1}{\text{gradient}})$ ✓ ε in the range 1.37(V) to 1.42(V) ✓	Readings must be within half a grid square ecf their value of gradient	3	3 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
08.4	Intercept ✓₁ Their value of intercept × their 08.3 ✓₂ OR <i>I</i> and <i>R</i> from obtained from graph ✓₁ Attempted evaluation using valid equation(s) ✓₂ <i>r</i> consistent with their values ✓₃	Accept 0.27 to 0.33 for newly read intercept OR Allow ecf for readings from 08.3 Expect to see answers in the range 0.37 to 0.47 (Ω)	3	3 × AO3
Total			9	

Question	Answers	Additional comments/Guidelines	Mark	AO
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13	A	0.6 V.	AO2
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Question	Answers	Additional comments/Guidelines	Mark	AO
02	comment about circuit when switch was open ✓ comment that closing the switch produces a current in circuit/cell AND cell has internal resistance ✓ conclusion that voltmeter reading decreases with relevant reason ✓	Condone 'battery' for 'cell'. MP1 e.g. voltmeter reading is the emf OR the emf is 1.53 V OR no current present, so no lost volts/energy loss. MP2: Reject idea that current causes internal resistance. Reject idea that there was any current when switch is open. MP3: e.g. 'terminal pd/voltage' decreases; idea of 'lost volts' in cell; pd exists across internal resistance. Allow explanation in terms of $\varepsilon = I(R + r)$ if terms defined. Allow explanation in terms of a potential divider circuit.	3	3 × AO2
Total			3	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
05.1	ratio of potential difference across R to current in R OWTTE ✓	Allow $R = \frac{V}{I}$ where V is pd across R and I is current in/through R. Condone “voltage” for pd. Reject any reference to pd per unit current.	1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
05.2	(non-ideal) ammeter has some resistance ✓₁ so: current in R/circuit is reduced ✓_{2a} use of $V = IR$ to explain V_2 less than V_1 ✓_{3a} OR pd occurs across ammeter ✓_{2b} potential divider argument for why V_2 is less than V_1 ✓_{3b} OR $V_1 = \text{emf}$ ✓_{2c} $V_2 = \text{emf} - \text{pd across ammeter}$ ✓_{3c}	Any reference to “internal resistance” must clearly relate to the ammeter. e.g. for ✓_{3a}: as R is constant, if I decreases, V_2 decreases according to $V=IR$ e.g. ✓_{3b}: emf is now shared between ammeter and R, so V_2 is less than V_1	3	1 × AO1 2 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.3	Yes because: no current occurs in the ideal voltmeter, or ammeter gives an accurate measurement for I in R OR voltmeter only measures pd across R , or V_2 is the accurate pd of R ✓	Condone idea that V_2 and I are a consistent pair of readings for R , or that adding ammeter leads to a (proportionate) decrease in V and in I	1	AO4
Total			5	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
08.1	uses parallel resistor formula or divides 2.40 by 2 with reason to show equivalent R of lamps is 1.20 ✓ adds 0.061 to 1.20 to get 1.26 ✓	MP2: internal resistance must be in Ω	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
08.2	determines total current using Ohm's law OR attempts to use relevant resistances in a potential divider equation ✓ 12 (V) ✓	MP1: Expect 10.151 A for 1.26 Ω ; and 9.846 A for 1.3 Ω MP1: may see $\frac{1.2}{1.261}$ or $\frac{0.061}{1.261}$, with rounded values of 1.261 Expect to see 11.8, 12.18 or 12.20 V	2	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
08.3	external/total R will increase so total current will decrease ✓ leading to fewer 'lost volts', so terminal pd increases ✓ OR load R will increase, so ratio of load R:internal r or load R:total R increases ✓ leading to larger proportion of emf across load, so terminal pd increases ✓		2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
08.4	Power in Fig. 12 is greater than in Fig. 11 because: current in X increases ✓_{1a} and refers to $P = I^2 R$ ✓_{2a} OR V increases ✓_{1b} and refers to $P = \frac{V^2}{R}$ ✓_{2b} OR <u>V and I</u> increase ✓_{1c} and refers to $P = VI$ ✓_{2c}	MP2 is contingent on MP1.	2	AO2
Total			8	

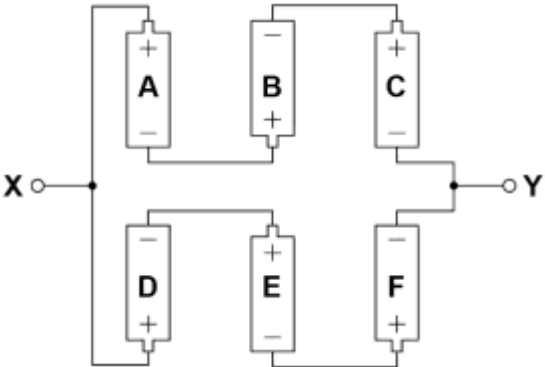
Question	Answers	Additional comments/Guidelines	Mark	AO
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11	C	$\text{A}^{-2} \text{kg m}^3 \text{s}^{-3}$	
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16	B	30 Ω	

Question	Answers	Additional comments/Guidelines	Mark	AO
03	Idea that (each) term is a work done per unit charge Idea that the LHS is energy input (per unit charge) by the cell and RHS is work done (per unit charge) in the cell + work done (per unit charge) in the rest of the circuit		2	AO1
Total			2	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
05.1	correct connections drawn ✓	expect: 	1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.2	Max 2: ✓✓ - Idea that the three cells D,E,F will run down more quickly or heat up more OR the current in DEF will increase - The total emf remains the same - The total internal resistance will double (because no current flows in A,B,C) - The current (drawn by the toy/between X and Y) will decrease OR the terminal pd will decrease	For MP2 and MP3, a simple calculation is insufficient. The internal resistance or emf must be compared to the previous one. Eg condone 'r is 0.75 ohm which is larger than before' For MP4, ignore any attempt to quantify Ignore any conclusions about whether the toy will stop 'working'	2	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
05.3	Any two of: ✓✓ - total emf (= 4.5 V) - total resistance (= 0.375 Ω) - use of $I = \frac{V}{R}$ 12 (A) ✓		3	AO1 AO2 AO3
Total			6	

Question	Answers	Additional comments/Guidelines	Mark	AO
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12	D	$\text{kg m}^3 \text{s}^{-3} \text{A}^{-2}$	AO1
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13	C	8 V	AO2
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15	B	increase	decrease	AO2
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Question	Answers	Additional comments/Guidelines	Mark	AO
04.1	Uses equation for A or determines A OR reads off ρ as $0.97 \times 10^{-7} \Omega \text{ m}$ ✓ Uses $\rho = \frac{RA}{l}$ ✓ 0.26 (m) ✓	MP1: $A = 2.545 \times 10^{-10} \text{ m}^2$. Condone power of ten (POT) error in A. MP2: Allow POT for ρ. Allow their incorrect A e.g. $1.02 \times 10^{-10} \text{ m}^2$ (from using 1.8×10^{-5} as r) MP3: Calculator value = 0.2597 m Allow 2 marks for 1.04 m (from $1.02 \times 10^{-10} \text{ m}^2$)	3	2 × AO2 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
04.2	Evidence of relevant method (may be seen on Fig. 5) ✓ $1.8 \times 10^{-7} (\Omega \text{ m})$ ✓	Accept for MP1: - determines gradient: allow 4.1×10^{-10} or 4.2×10^{-10}. Condone POT error. - uses $y = mx + c$ - use of similar triangles - use of $y - y_1 = m(x - x_1)$	2	AO4
Total			5	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
05.1	Potential divider argument given OR calculates current as 0.0136 A ✓ 660 (Ω) from some relevant working ✓	MP1: e.g. ratio of pds (at max R) is 3:9, so ratio of resistances is 1:3 MP2: 0.0136 A gives 662 Ω	2	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
05.2	There will now be a pd across internal resistance _{1a}✓ Idea that emf is now shared across internal resistance and load resistance/R/220 Ω _{2a}✓ OR Current in circuit will reduce _{1b}✓ Uses 220 Ω in $V=IR$ to conclude that reading will be <12 V _{2b}✓	MP1a: Condone 'lost volts' MP2a: May see "terminal pd = emf – lost volts" used. Allow potential divider arguments.	2	AO1
Total			4	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
09.1	Electron transfers energy to a mercury atom by collision ✓ Collision excites/ionises the atom OR moves electron to higher (energy) level or ejects electron from atom ✓ (mercury) Atom/ion de-excites, emitting (UV) photon ✓	Accept “Hg” for “mercury”. A reference to “collision” is needed only once if both MP1 and MP2 are awarded.	3	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
09.2	Substitution into $E = hf$ ✓ Divides 42×10^{-3} by their E ✓ 8.3×10^{16} or 8.4×10^{16} ✓	MP1: Expect 5.039×10^{-19} (J) MP2: Condone POT error in power MP3: Calculator value = 8.3353×10^{16}	3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
09.3	Uses $P = VI$ with 3.4 V OR reads off I as 19 mA ✓ 0.65 ✓	MP1: Expect $P = 64.6$ mW MP2: Allow 0.63 to 0.67, or 63% to 67%.	2	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
09.4	Uses $V = IR$ to get pd across fixed R _{1a}✓ Reads off, and adds, pds for LEDs _{2a}✓ Determines resistance of each LED _{1b}✓ Finds total resistance (with $V = IR$) _{2b}✓ Value that rounds to 9.0 (V) ₃✓	MP1a: Expect 1.161 V. Condone POT error for current. MP2a: Expect $V_{\text{red}} = 2.05 \text{ V}$; $V_{\text{green}} = 2.70 \text{ V}$; $V_{\text{blue}} = 3.05 \text{ V}$. Condone 1 misreading beyond $\pm 0.05 \text{ V}$ of expected value. MP1b: Expect $R_{\text{red}} = 47.7 \Omega$; $V_{\text{green}} = 62.8 \Omega$; $V_{\text{blue}} = 70.9 \Omega$. Condone 1 miscalculation. MP2b: Expect $181.4 + 27 = 208.4 \Omega$ MP3: Reject a 1 sf answer.	3	AO3
Total			11	

Question	Answers	Additional comments/Guidelines	Mark	AO
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24	D	$5I_1 = 3I_2 + 2I_3$	AO3
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Question	Answers	Additional comments/Guidelines	Mark	AO
06.1	Read-off = 1.6(0) (V) ✓		1	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
06.2	Gradient determined using points far apart ✓ Gradient negative, resistance positive ✓ 0.19 (Ω) ✓	Expect to see eg $\frac{1.6}{8.6}$	3	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
06.3	Lamp with B is brighter because: lower internal resistance of B means less lost volts ✓ (so) higher terminal pd ✓ OR Lamp with B is brighter because: lower total resistance ✓ (so) greater current ✓		2	AO2

Total			6	
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Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
11.1	Read-off from Figure 15 for 37 °C (7.5 to 8 k Ω) ✓ Use of potential divider equation or other valid method ✓ Answer that rounds to 2100–2200 Ω ✓		3	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
11.2	When resistance of R is increased, it has a larger share of the voltage (for any particular temperature) OR the current in the circuit decreases ✓ The voltage of the thermistor is decreased (for the same temperature) ✓ The threshold voltage is reached at a lower temperature ✓		3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
11.3	Idea that the light can turn on at some other threshold temperature so the chosen temperature for the bath water can be adjusted to be warmer or cooler ✓		1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
11.4	Very high resistance so it does not affect the voltages in the other components of the circuit (ie the potential divider shown in Figure 14) ✓		1	AO3
Total			8	

Question	Key	Answer	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
05.1	Any two from: ✓✓ • use of $V = IR$ to find I • reads off V for their I • adds their V to 2.6 V = 3.8 to 4.0 (V) ✓	expect 0.055 A expect 1.2 to 1.4 V	3	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
05.2	A tangent is drawn at the origin and the voltage where the curve leaves the tangent is read off ✓ $V_1 = 0$ (V), $V_2 = 0.80$ to 0.95 (V) ✓	Allow 1sf Do not allow 1V for MP2	2	AO3
Total			5	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
06.1	At least two from: ✓ - metal bar has larger cross-sectional area than the wire - metal bar has shorter length than the wire - metal bar is made of a lower resistivity metal than the wire reference to resistivity equation OR both corresponding proportionalities from the equation ✓	For MP2, must be standard symbols from data booklet or symbols defined	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.2	Correct expression for length of wire ($= \pi \times 0.045 \times 16 \times 10 = 22.6 \text{ m}$) ✓ Correct expression for cross-sectional area of wire = $\frac{\pi}{4} \times (0.91 \times 10^{-3})^2 (= 6.5 \times 10^{-7} \text{ m}^2)$ ✓ Use of resistivity equation $\rho = \frac{RA}{l}$ (with their A and their l) ✓ $5.3 \times 10^{-7} (\Omega \text{ m})$ ✓	mp1 and mp2 and mp3 can be combined Allow POT error for MP2 and MP3 only Allow 3 marks total for final answer with only POT error	4	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.3	Idea that the other components in the ohmmeter circuit have resistance or contact resistance ✓	Accept idea that even when the slider is at B , some of the long wire is in the circuit Ignore reference to ohmmeter having its own resistance	1	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
06.4	Use of potential divider method OR Expression for current using total resistance then use of $V = IR$ with 18.3 ohm ✓ $= 1.5(2) \text{ (V)} \checkmark$	Expect to see 0.083 A	2	AO2
Total			9	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO																					
10.1	Correct values in Table 1 to 3 sf ✓		2	1 × AO2 1 × AO3																					
	Their values correctly plotted ✓																								
	<table><tr><td>R / Ω</td><td>I / A</td><td>$\frac{1}{I} / A^{-1}$</td></tr><tr><td>4.0</td><td>0.991</td><td>1.01</td></tr><tr><td>5.0</td><td>0.872</td><td>1.15</td></tr><tr><td>6.0</td><td>0.781</td><td>1.28</td></tr><tr><td>7.0</td><td>0.680</td><td>1.47</td></tr><tr><td>8.0</td><td>0.611</td><td>1.64</td></tr><tr><td>9.0</td><td>0.560</td><td>1.79</td></tr></table>				R / Ω	I / A	$\frac{1}{I} / A^{-1}$	4.0	0.991	1.01	5.0	0.872	1.15	6.0	0.781	1.28	7.0	0.680	1.47	8.0	0.611	1.64	9.0	0.560	1.79
	R / Ω				I / A	$\frac{1}{I} / A^{-1}$																			
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Question	Answers	Additional comments/Guidelines	Mark	AO
10.2	Line of best fit, extended to the y-axis ✓		1	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
10.3	Rearranges $\varepsilon = I(R + r)$ to make R the subject ✓ and links to $y = mx + c$ ✓	Expect to see $R = \varepsilon \times \frac{1}{I} - r$ Eg by aligning each term vertically	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
10.4	Determines gradient from large triangle ✓ $\varepsilon =$ between 6.0 and 6.4 (V) ✓ $r =$ between 2.0 and 2.6 (Ω) ✓	Two read-offs must be at least half the line apart. One can be y-intercept MP2 and MP3 are quality marks	3	AO2
Total			8	

Question	Answers	Additional comments/Guidelines	Mark	AO
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15	D	decreases increases	AO1
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16	D	$\frac{3\varepsilon R}{3R + r}$	AO1
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